

Session: Learning From Data

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DSA 8401: Applied Machine Learning
Master in Data Science and Analytics



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Objectives

- Obtain unbiased estimates of a model's performance
- Diagnose common issues in machine learning algorithms
- Tune machine learning models
- Evaluate predictive models using different metrics

All the models are wrong but some are useful.

Summary

Basics

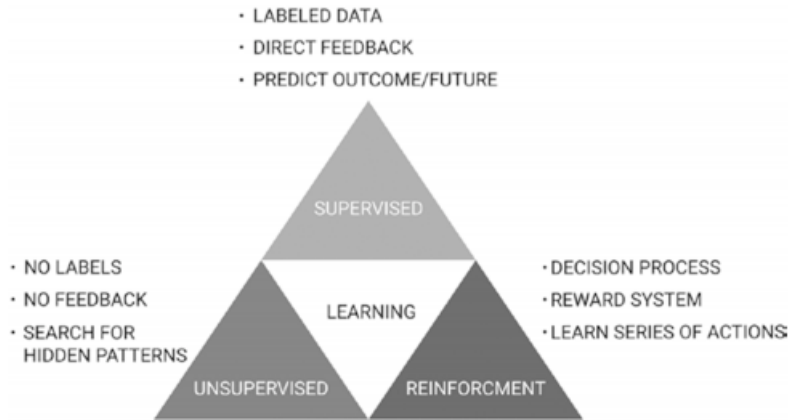
Supervised Algorithms

Model Selection and Evaluation

Metrics to assess model performance

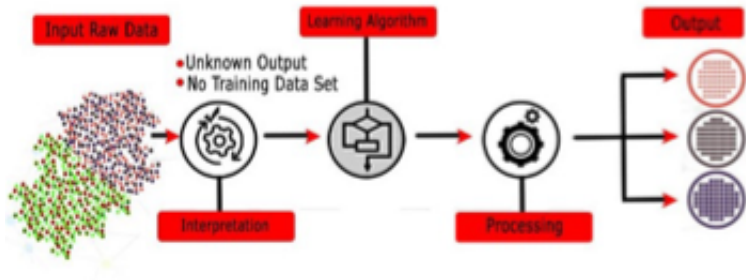
Types of Machine Learning?

Supervised



Types of Machine Learning?

Unsupervised Learning



Types of Machine Learning?

Reinforcement Learning



Definition of a ML project

Tom Mitchell's definition of machine learning:

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."

In simpler terms, a machine learning algorithm learns from data (experience) to improve its ability to perform a specific task (e.g., recognizing images, predicting stock prices).

How ML algorithm works?

Examples of tasks and experience along with performance metrics.

Problem	Task	Performance	Experience
Image Recognition	Recognizing a person's weight from an image	Accuracy in weight prediction	Dataset with images of a person and their weight
Robotic Arm for Medication Administration	Moving the correct medication to the correct patient container	Percentage of medications dispensed correctly	Training examples from real-world experience
Disease Risk Prediction	Diagnosing the probability of developing Type 2 Diabetes	Percentage of correct diagnoses	Dataset with clinical records of patients and real-world experience

Summary

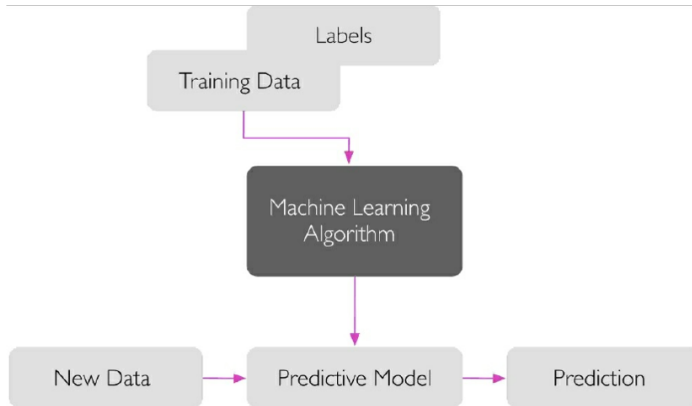
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Supervised Learning



Supervised Learning

Supervised learning problems can be classified into two types:

Classification: To predict the outcome based on a training data set where the **output variable is in the form of distinct categories**. Models are created using training data in the form of labeled data. An example of classification is labeling (in a real clinical application, this would be the diagnosis) someone as sick or healthy based on a set of symptoms.

Regression: Very similar to classification. The only difference between classification and regression is that regression is the result of a given sample where the **output variable is in the form of real values** (i.e., temperature as a value, instead of a classification of cold or hot). Examples include blood O_2 , body temperature, ...

Supervised Learning

Mathematically

Supervised learning is a computational task to infer a function from labeled training data.

$$Y = f(X)$$

which can be understood as

$$\textit{Output} = \textit{function}(\textit{Input})$$

The inputs X and outputs y are usually referred to as variables or attributes and are typically in vector format. f refers to the function (model) we are trying to infer from the input/output.

Supervised Learning

"No Free Lunch" Theorem

This theorem states there is no one-size-fits-all algorithm that performs best across all possible problems.

1. No single learning algorithm will outperform all others on every possible task or dataset.
2. Performance differences between algorithms depend on the specific problem being solved, including the nature of the data and the assumptions made by the algorithm.

The theorem emphasizes the importance of selecting the right algorithm for the particular problem you're dealing with, rather than expecting one algorithm to always work best.

Supervised Algorithms

- Linear and logistic regression
- Support Vector Machines (SVM)
- Decision trees and Ensemble methods: Bagging (Random Forest), Boosting, ...
- Artificial Neural Networks

Summary

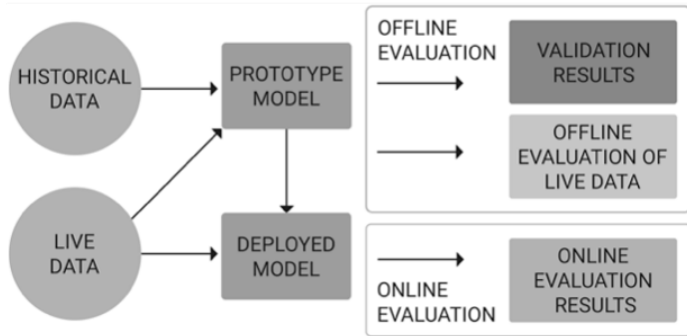
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Data flow for model development



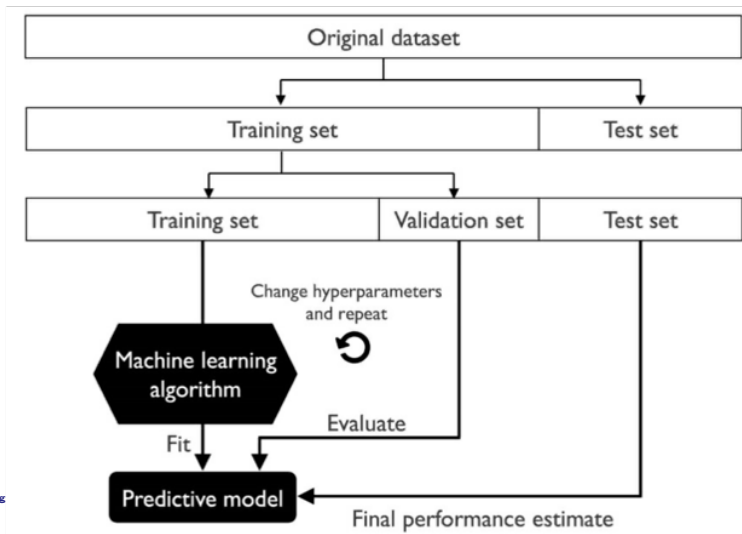
How to choose an Algorithm

Steps

1. Select the features to be used and collect training samples.
2. Choose a performance metric.
3. Choose a classifier and optimization algorithm.
4. Evaluate the model's performance.
5. Fine-tune the algorithm.

Model Performance

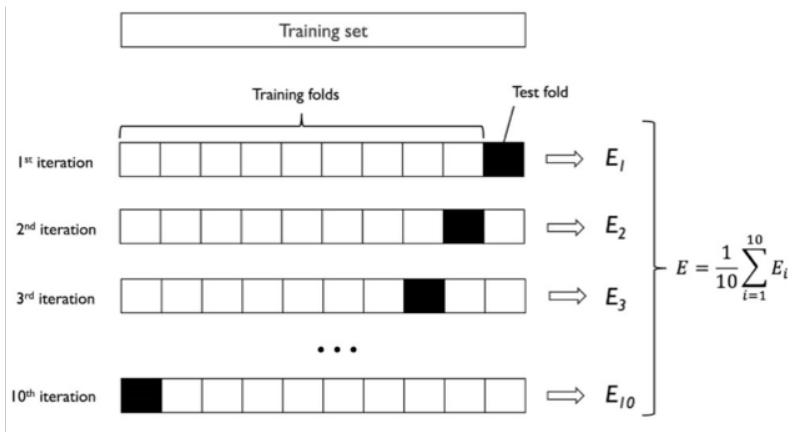
Holdout Method



Model Performance

Cross Validation

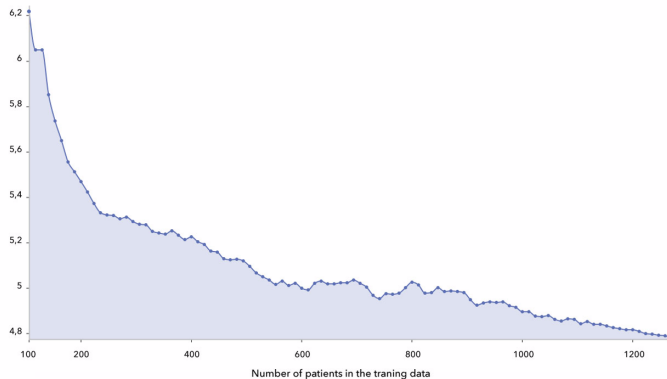
Suitable when there are few data



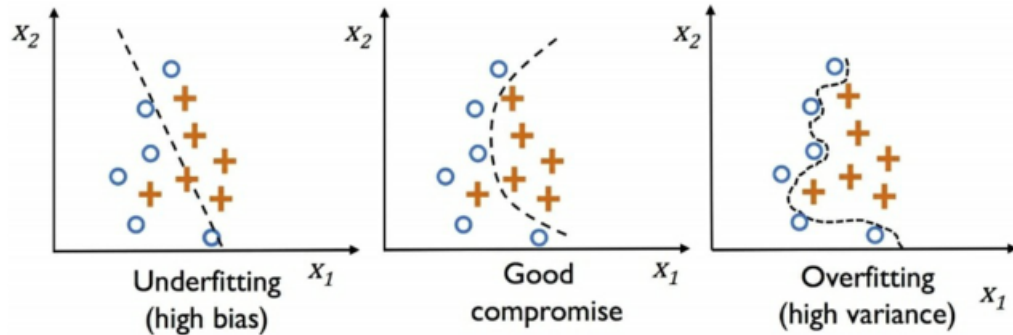
How much data we need?

The more the better

Mean Absolute Error

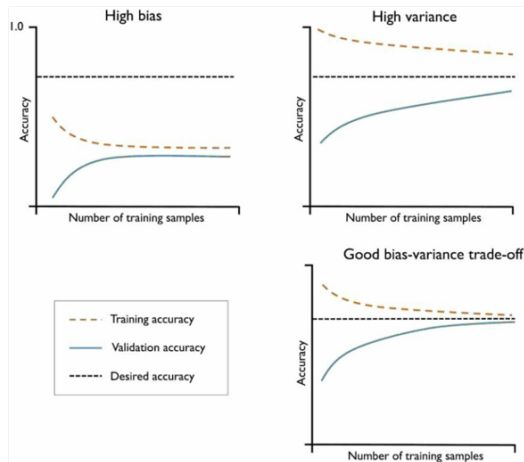


Problem: Overfitting



- A model can work well with training set but not with test set. **Do not generalize**

Learning curves



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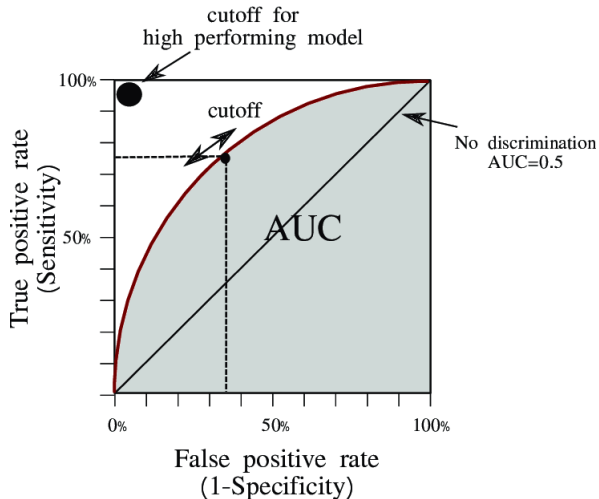
Metrics to assess model preformance

Confusion matrix and related metrics

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Visualizing Performance: ROC and AUC

Receiver Operating Characteristic (ROC)



Data Unbalanced

- Common problem in real world
- Predicting always the dominant class we get good accuracy but it is useless
- The optimization methods used by the algorithm bias the results to the dominant class.
- The solution is
 - to reduce the data of the dominant class
 - to increase the data of the minority class
 - to create synthetic data

Conclusions

- When using supervised methods, it is very important that the models we obtain generalize well and do not suffer from overfitting
- To avoid this, we use two-step training (holdout method) and ensure that there is enough data, while also trying to prevent the data from being imbalanced.
- We tend to use accuracy because everyone has a clear idea of what it means
- More appropriate metrics, such as sensitivity and precision, may seem unfamiliar, but they work better for certain problems, like unbalanced classification tasks